

Sri Chaitanya IIT Academy.,India.

A.P T.S KARNATAKA TAMILNADU MAHARASTRA DELHI RANCHI

A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **Paper -1(Adv-2020-P1-Model)** **Date: 26-04-2026**

Time: 09.00Am to 12.00Pm **GTA-27** **Max. Marks: 198**

26-04-2026_Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT_Jee-Adv(2020-P1)_GTA-27_Syllabus,

PHYSICS : TOTAL SYLLABUS

CHEMISTRY : TOTAL SYLLABUS

MATHEMATICS : TOTAL SYLLABUS

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Name of the Student: _____ **H.T. NO:** | | | | | | | | |

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JEE-ADVANCE-2020-P1-Model

Time: 3:00Hour's **IMPORTANT INSTRUCTIONS** **Max Marks: 198**

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total mark
Sec – I(Q.N : 1 – 6)	Questions with Single Correct Choice	3	-1	6	18
Sec – II(Q.N : 7 – 12)	**Questions with Multiple Correct Choice +1 partial marks**	4	-2	6	24
Sec – III(Q.N : 13 – 18)	Questions with Numerical Value Answer Type	4	0	6	24
Total				**18**	**66**

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 19 – 24)	Questions with Single Correct Choice	3	-1	6	18
Sec – II(Q.N : 25 – 30)	**Questions with Multiple Correct Choice +1 partial marks**	4	-2	6	24
Sec – III(Q.N : 31 – 36)	Questions with Numerical Value Answer Type	4	0	6	24
Total				**18**	**66**

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 37 – 42)	Questions with Single Correct Choice	3	-1	6	18
Sec – II(Q.N : 43 – 48)	**Questions with Multiple Correct Choice +1 partial marks**	4	-2	6	24
Sec – III(Q.N : 49 – 54)	Questions with Numerical Value Answer Type	4	0	6	24
Total				**18**	**66**

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IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
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JEE Advanced 2023 VAVILALA CHIDVILAS REDDY RANK 1 341/360
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PHYSICS **Max Marks: 66**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

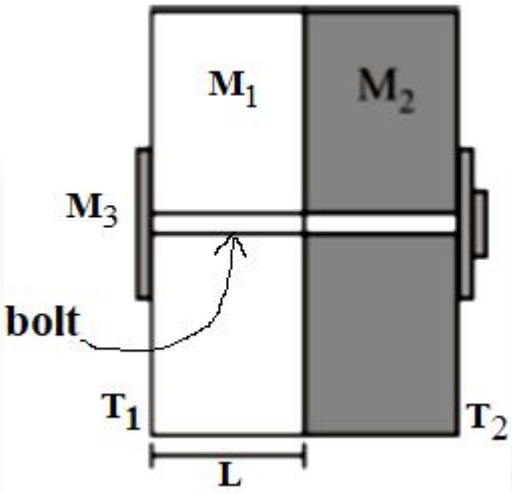
This section contains **SIX** (06) questions.
• Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
• For each question, choose the option corresponding to the correct answer.
• Answer to each question will be evaluated **according to the following marking scheme:**
Full Marks: +3 If ONLY the correct option is chosen;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: -1 In all other cases

1. An accurate pendulum clock is mounted on the ground floor of a high building. How much time will it lose or gain in one day if it is transferred to top storey of a building which is h = 200 m higher than the ground floor? Radius of earth is $6.4\times10^{6}m$.
A) It will lose 6.2 s B) It will lose 2.7 s
C) It will gain 5.2 s D) It will gain 1.6 s

2. A vertical glass jar is filled with water at $10^{\circ}C$. It has one thermometer at the top and another at the bottom. The central region of the jar is gradually cooled. It is found that the bottom thermometer reads $4^{\circ}C$ earlier than the top thermometer and the top thermometer reads $0^{\circ}C$ earlier than the bottom thermometer. This happens because
A) the top thermometer is faulty
B) the bottom thermometer is faulty
C) density of water is maximum at $4^{\circ}C$
D) density of water is minimum at $0^{\circ}C$

3. Two different materials M_1, M_2 of equal dimensions (length L and uniform cross–sectional area A) having thermal conductivities k_1 and k_2 respectively are connected together through a bolt M_3. The bolt (length = 2L, area = A/n) is of another material with thermal conductivity k_3 such that n > > 1. The left sides of the materials M_1 and M_3 are kept at the temperature T_1 whereas the right sides of the materials M_2 and M_3 are kept at the temperature T_2 as shown in the figure below such that $T_1 > T_2$. Assume that the bolt (M_3) is thermally insulated from materials M_1 and M_2. The equivalent thermal resistance of this system is

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A) $\frac{L(k_1+k_2)}{Ak_1k_2}$ B) $\frac{L}{(k_1+k_2)A}$

C) $\frac{L(k_3+2n(k_1+k_2))}{A(k_1+k_2)nk_3}$ D) $\frac{2nL(k_1+k_2)}{A(2nk_1k_2+k_2k_3+k_3k_1)}$

4. When we press the nozzle of a deodorant, we feel cooling effect. This is due to

i) The liquid is cooled at the factory and is distributed to us in adiabatic containers, so the liquid remains cold.

ii) The liquid coming out as small droplets so has more surface area, hence its internal energy is utilized to create the surface energy.

iii) The liquid comes out through a tube which is near the bottom of the container. The colder liquid is at the base due to higher density.

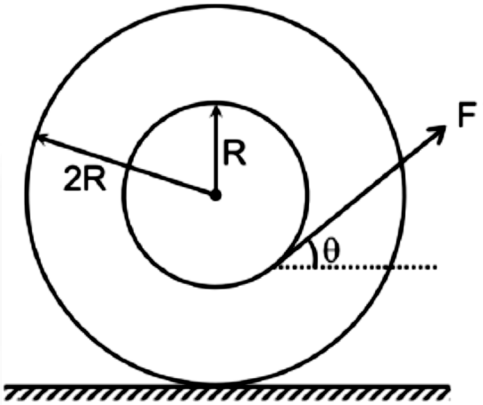
iv) The liquid is coming out as small droplets so has more surface area, hence it can absorb heat quickly from our body.

A) Only i and ii are correct B) Only ii and iv are correct

C) Only i, ii and iii are correct D) Only ii, iii and iv are correct

5. A spool of mass 'm' has inner and outer radius R and 2R respectively. A thread is wound on the inner disc and its free end is pulled with a force F at an angle θ with the horizontal in the direction as shown in the figure. Assume no slipping condition between thread, spool and between spool and ground and $I_{cm} = 2mR^2$, then

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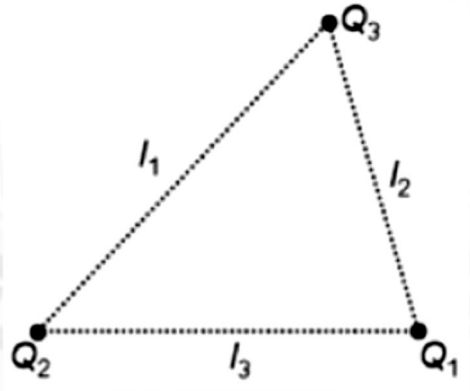
A) for $0 \le \theta \le 90^\circ$, frictional force on spool at ground acts leftward only.

B) for $0 \le \theta \le 60^\circ$, friction on spool at ground acts leftward and for $60^\circ \le \theta \le 90^\circ$, it acts rightward.

C) for a certain θ between 0° and 90° friction becomes zero.

D) none of the above are correct

6. Figure shows three small positively charged particles having charges Q_1, Q_2 and Q_3 and masses m_1, m_2 and m_3 respectively are placed at the vertices of a triangle having sides ℓ_1, ℓ_2, ℓ_3. When the charges are released and allowed to move freely under mutual electrostatic forces, it was observed that they all move along different straight lines. If $\frac{Q_1}{m_1} : \frac{Q_2}{m_2} : \frac{Q_3}{m_3} = \frac{1}{27} : \frac{1}{8} : \frac{1}{64}$, find $\ell_1 : \ell_2 : \ell_3$.



A) 27 : 8 : 64 B) 3 : 2 : 4 C) 4 : 3 : 2 D) 2 : 6 : 9

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

- This section contains **SIX** (06) questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).

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• Answer to each question will be evaluated **according to the following marking scheme**:
Full Marks: +4 If only (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but ONLY three options are chosen;
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct;
Partial Marks: +1 If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: -2 In all other cases

7. Two small identical smooth blocks A an B are free to slide on a frozen lake. They are joined together by a light elastic rope of length $\sqrt{2}L$ which has the property that it stretches very little when the rope becomes taut. At time $t=0$, A is at rest at $x=y=0$ and B is at $x=L,\ y=0$ moving in the positive y–direction with speed V. Then

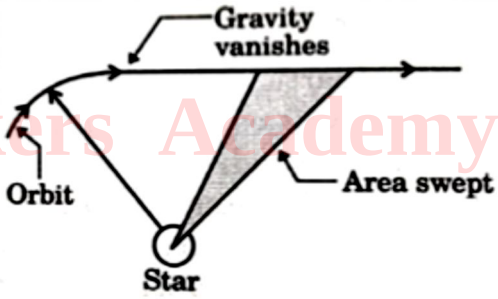
A) positions of A and B at $t=\dfrac{2L}{V}$ are $\left(\dfrac{L}{2},\dfrac{L}{2}\right)$ and $\left(\dfrac{L}{2},\dfrac{3L}{2}\right)$ respectively.

B) positions of A and B at $t=\dfrac{100L}{V}$ are $(L,50L)$ and $(0,50L)$ respectively.

C) at $t=\dfrac{2L}{V}$, velocities of A and B are $\overrightarrow{V_A}=\dfrac{V}{2}\left(\hat{i}+\hat{j}\right)$ and $\overrightarrow{V_B}=\dfrac{V}{2}\left(-\hat{i}+\hat{j}\right)$.

D) at $t=\dfrac{100L}{V}$, velocities of A and B are $\overrightarrow{V_A}=V\hat{j}$ and $\overrightarrow{V_B}=\vec{O}$.

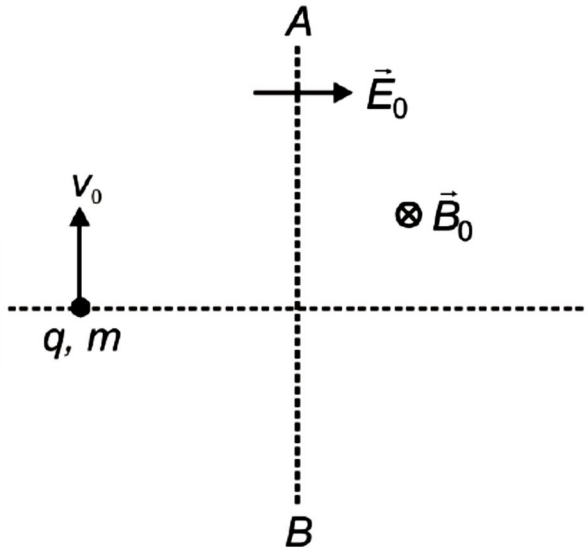
8. A planet is orbiting a star. For no apparent reason the star's gravity suddenly vanishes, after which planet moves in a straight line. Mark the correct statement(s):



A) Newton's first law is obeyed on planet after gravity vanishes also.
B) Kepler's law of areas is obeyed only till the planet is in gravity of star.
C) Kepler's law of areas is obeyed even after gravity vanishes.
D) Angular momentum of planet about centre of star is conserved through out its motion.

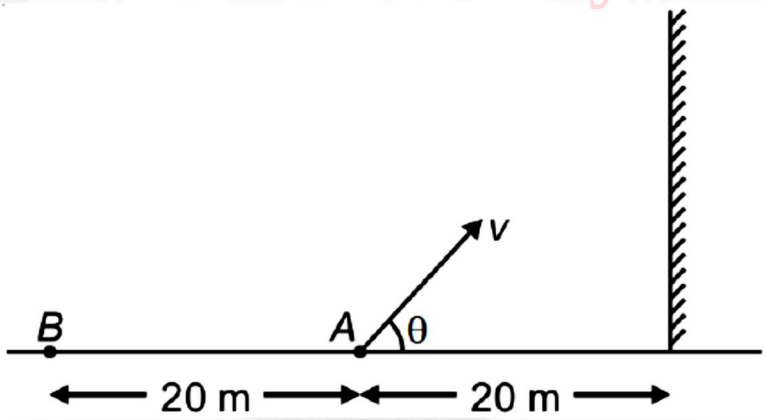
9. A uniform electric field $\vec{E}_0$ is present horizontally along the paper throughout the region but uniform magnetic field B_0 is present perpendicular to the plane inward, only right to the line AB as shown. A charged particle (q, m) is projected vertically upward with velocity v_0 and crosses the line AB after time t_0. Find the speed of projection v_0 if particle moves with constant velocity after t_0. (Given that $qE_0=mg$)

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A) gt_0 B) $\frac{E_0}{B_0}$ C) $\frac{2E_0}{B_0}$ D) $2gt_0$

10. In the given figure, a stationary boy (at point A) throws a ball towards a fixed vertical wall which is 20 m away from him. The ball hits the wall elastically and lands at a point B 20 m behind the boy. The total time elapsed is T_1. The boy again throws the ball with same speed from same point A towards the wall, and after collision with the wall, ball lands at same point B. But this time, total time elapsed is T_2. If $|T_2 - T_1| = 1s$, choose the correct options. $\left(g = 10\ m/s^2\right)$



A) Speed of throw of ball is 20 m/s

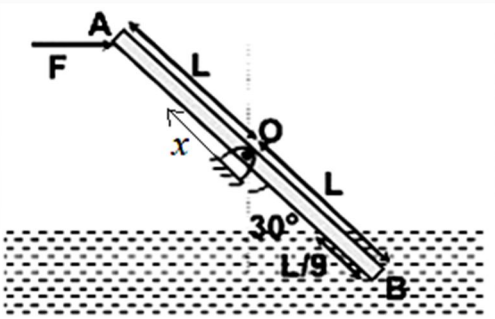
B) Speed of throw of ball is 25 m/s

C) The difference in their angle of projections is nearly 16°

D) The difference in the height of the points where ball hit the wall is nearly 10 m.

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11. A rod of length 2L and cross section area A_0 is kept in a liquid of density ρ and pivoted at point O in a vertical plane. The rod makes angle 30° with the vertical. The mass density of rod linearly increases from O on both sides as given by equation $\lambda = \lambda_0\left(1+\frac{x}{L}\right)$, where λ_0 is constant. Rod is held in vertical plane by applying a force at end A in horizontal direction and $\frac{L}{9}$ length of rod is in the liquid. Then



A) centre of mass of rod OA is at a distance $\frac{4L}{9}$ from point A.

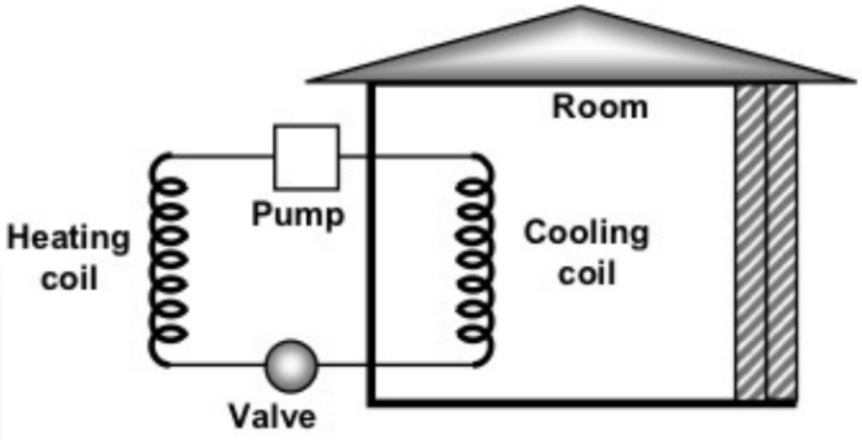
B) mass of rod AB is $\frac{3\lambda_0 L}{2}$

C) applied force at end A is $\frac{17}{162\sqrt{3}}LA_0\rho g$

D) buoyant force will act at the centre of mass of rod inside the liquid.

12. A room air conditioner is modified as a heat engine run in reverse. An amount of heat Q_{in} is absorbed from the room at a temperature T_L by the cooling coils which contains a working gas; this gas is compressed adiabatically to a temperature T_H; then the gas is compressed isothermally in a coil outside of the house, giving off an amount of heat Q_{out} ; the gas expands adiabatically back to a temperature T_L and the cycle is repeatd. An amount of energy W is input into the system in every cycle through an electric pump. This model describes the practical air conditioner with the best possible efficiency.
The outside air temperature is T_H and the inside air temperature is T_L. The air conditioner unit consumes electric power P. Assume that the air is dry sufficiently so that no condensation of water occurs in the cooling coils of the air conditioner. Which of the following may be the rate at which heat is removed from the room?

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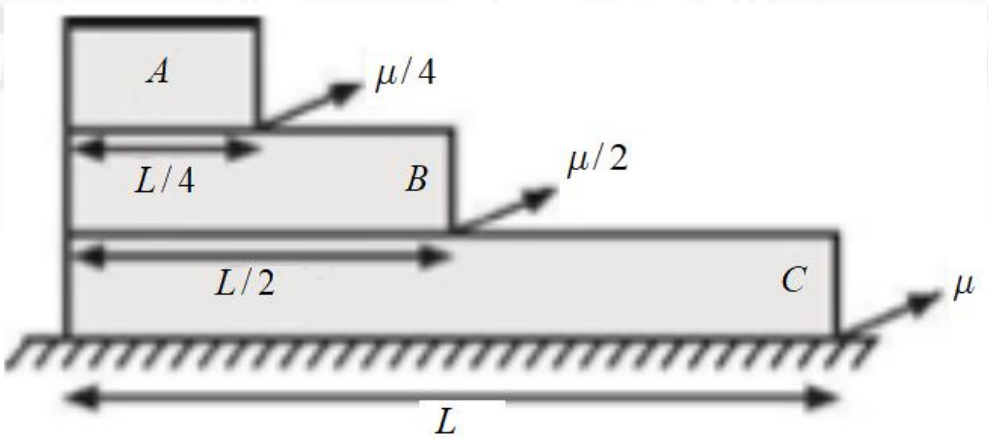


A) $\frac{P}{2}\left(\frac{T_L}{T_H - T_L}\right)$ B) $\frac{3P}{4}\left(\frac{T_L}{T_H - T_L}\right)$ C) $P\left(\frac{T_L}{T_H - T_L}\right)$ D) $\frac{3P}{2}\left(\frac{T_L}{T_H - T_L}\right)$

SECTION – III
(NUMERICAL VALUE TYPE)

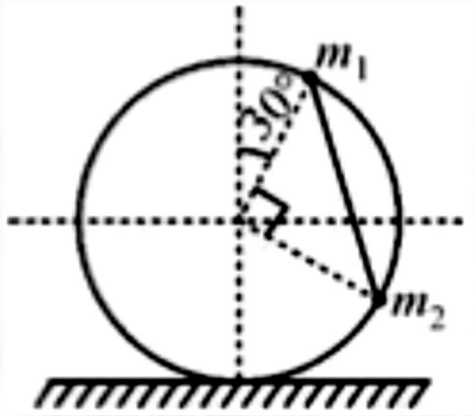
• This section contains **SIX** (06) questions. The answer to each question is a **NUMERICAL VALUE.**
• For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer. **If** the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.
• Answer to each question will be evaluated **according to the following marking scheme**:
Full Marks: +4 **If** ONLY the correct numerical value is entered;
Zero Marks: 0 In all other cases

13. Three blocks A, B and C of masses $m, \frac{m}{2}$ and m of different densities and dimensions are placed over each other as shown in the figure. The coefficients of friction are shown. Initially, left ends of blocks are placed in a vertical line and are made to move towards right with same velocity. Find the time (in sec) taken by the upper block A to topple from the middle block B. Assume that blocks B and C don't stop sliding before A topples from B. (Given L = 36 m, $\mu = 0.4$ and $g = 10\,m/s^2$)



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14. Two beads of masses m_1 and m_2 are connected to each other by a massless string. Both the beads are free to move on a fixed smooth circular loop in vertical plane. At an instant, they are in the position shown, subtend angle 90° at centre and system is released from rest. If the tension in string immediately after release is $T=\dfrac{\left(\sqrt{3}-1\right)m_1m_2g}{\sqrt{k}\left(m_1+m_2\right)}$, find k.

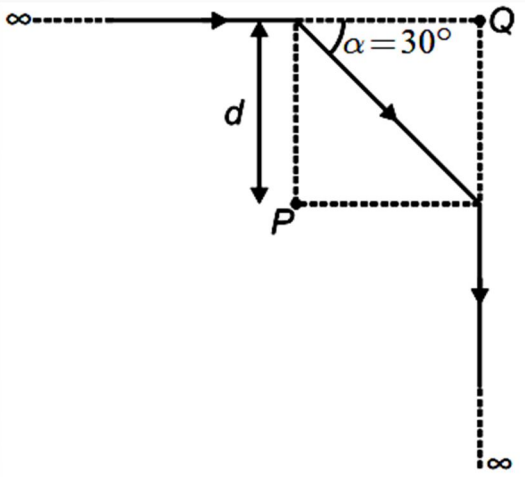


15. A cubical container of side a, wall thickness $x\left(x<<a\right)$ and thermal conductivity K is suspended in air and filled with n moles of a diatomic gas (adiabatic exponent = γ) in a room. Temperature of outer surface of container is maintained at T_0. If at $t=0$, the gas temperature is $T_1\left(T_1>T_0\right)$, find its temperature T as a function of time t. Assume the heat is conducted through all the walls of container. If $T=T_0+\left(T_1-T_0\right)e^{\dfrac{bKa^2\left(\gamma-1\right)}{nRx}}$, find b.

16. Sound waves of frequency 16 kHz are emitted by two coherent point sources of sound placed (symmetrically) 2 m apart at the center of a circular train track of large radius. A person in the train observes 2 maxima per second (on an average) when the train is running at a speed of 72 km/hr. Calculate the length of the circular track in km. (Velocity of sound in air 320 m/s)

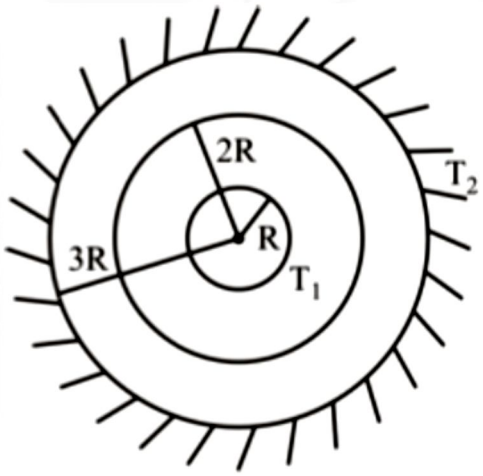
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17. For the given segment of current–carrying wire, the ratio of magnitudes of magnetic field strengths at point P to the point Q as shown is $\frac{B_P}{B_Q} =$ _____



18. Three concentric thin spherical shells are shown in figure. Outer most sphere can't radiate into outer space. The inner most and the outer most shells are maintained at T_1K and T_2K respectively. Assume the three shells behave as black bodies. If the steady state temperature of the middle shell is $\left(\frac{T_1^4}{x}+\frac{T_2^4}{y}\right)^{1/4}$, value of $\frac{x}{y}$ is _____

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CHEMISTRY **Max Marks: 66**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains **SIX** (06) questions.
• Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
• For each question, choose the option corresponding to the correct answer.
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Negative Marks: -1 In all other cases

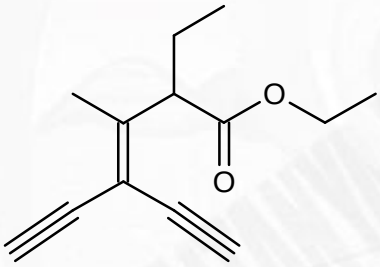
19. Which compound produces a coloured solution when placed in water under nitrogen atmosphere?

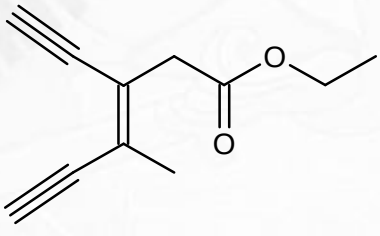
A) $CuC\ell_2$ B) $AgC\ell$ C) $ZnC\ell_2$ D) $Cu_2C\ell_2$

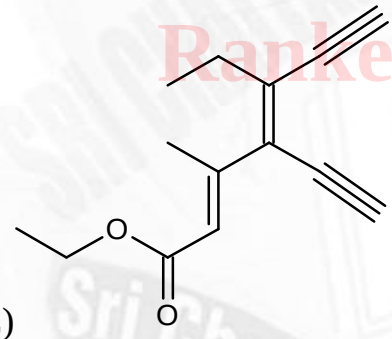
20. The characters of LUMO of CN^- and O_2 respectively, are

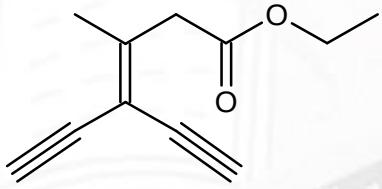
A) σ_g and π_u B) π_g and σ_u C) π_u and σ_u D) σ_u and π_g

21. The structure of ethyl 4-ethynyl-3-methylhex-3-en-5-ynoate is

A) 

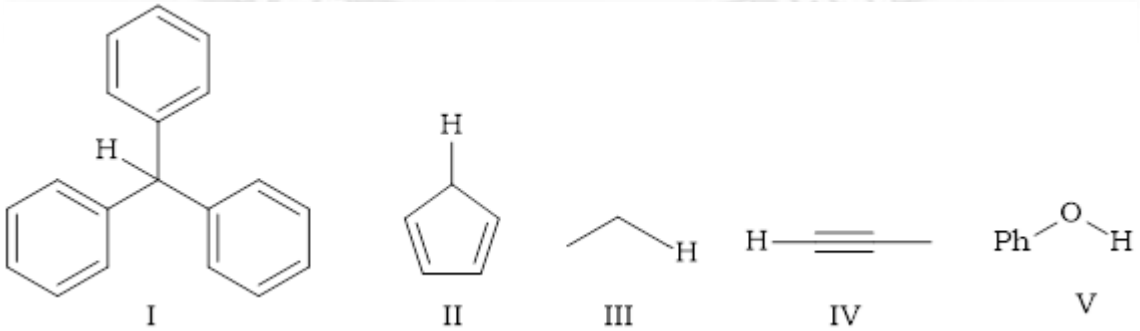
B) 

C) 

D) 

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22. With respect to compounds I-V choose the incorrect statement:



I II III IV V

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A) Acidic character of IV is due to higher % s-character of carbon.

B) II is more acidic than I due to aromatic stabilization of its conjugate base.

C) The order of acidic character of these compounds is II>V>IV>I>III

D) III becomes more acidic when it has a NO_2 substituent on any one the carbon atoms.

23. The Latimer diagram for nitrogen in acidic medium is given below:

$$NO_3^- \xrightarrow{+0.96V} NO_2 \xrightarrow{+1.04V} NO \xrightarrow{+0.96V} \frac{1}{2}N_2O \xrightarrow{+1.77V} \frac{1}{2}N_2$$

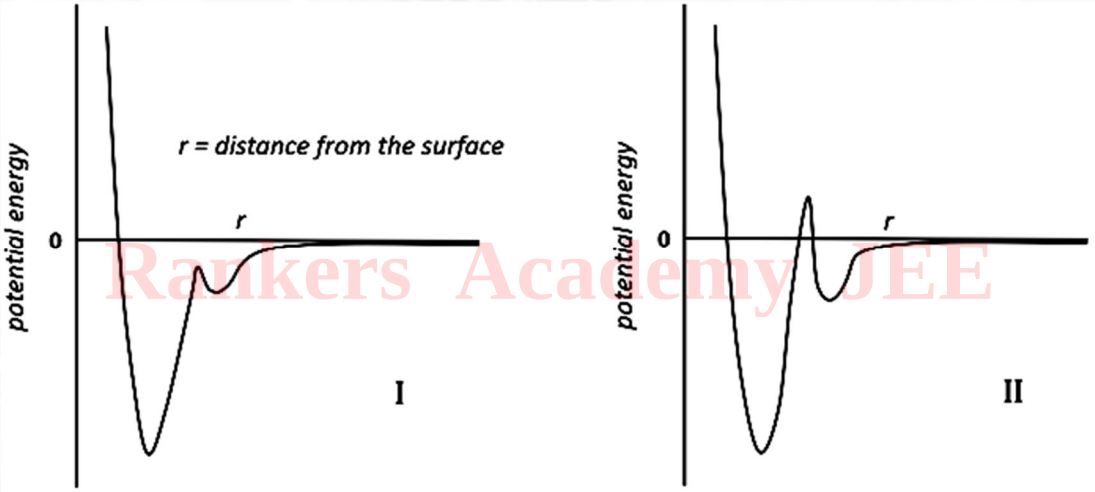
The numbers represent standard reduction potentials (in V).

The value of the standard reduction potential E° for the overall reaction

$NO_3^- + 4H^+ + 3e \longrightarrow NO + 2H_2O$ is closest to:

A) 2 V B) 1.01 V C) 1.03 V D) 0.67 V

24. Consider the potential energy profile diagram for the chemisorption of $A_{2(g)}$ on a metal surface by two different processes I and II:



Which among the following statement is **INCORRECT**?

A) In process I, the precursor state transforms into the chemisorbed state without any activation energy.

B) Process II involves activated adsorption and therefore shows strong temperature dependence.

C) Under identical conditions, process II is generally faster than process I.

D) At low temperatures, adsorption through process I is more favourable than through process II.

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SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

- This section contains **SIX** (06) questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated **according to the following marking scheme**:

Full Marks: +4 If only (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but ONLY three options are chosen;
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct;
Partial Marks: +1 If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: -2 In all other cases

25. Identify the **CORRECT** statement(s) for the adduct $H_3B \cdot CO$

A) sp^2 hybridized orbital of boron accept the lone pair of CO.

B) Formal charge on carbon is +1.

C) The shape of the adduct is tetrahedral around boron atom.

D) Formal charge present on boron atom is –1.

26. Which of the following are **CORRECT** regarding to the extraction of metal from its ore?

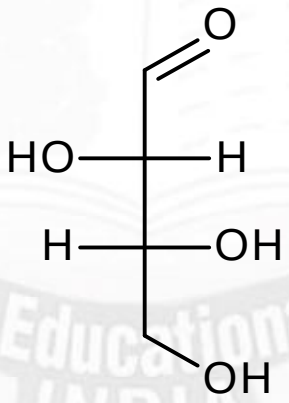
A) Generally slag is having lower density than molten metal.

B) Flux is used mainly to catalyze reduction.

C) Leaching is one of the process employed in the extraction of Aluminium.

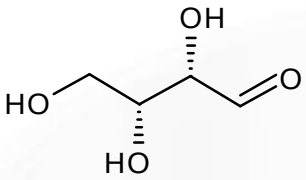
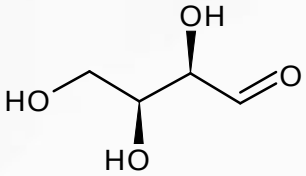
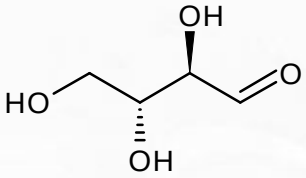
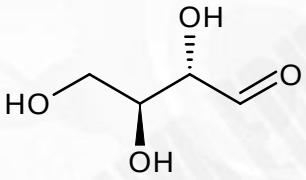
D) The main reducing agent in the reduction of iron is CO.

27. The Fischer projection of D-Threose is shown below:

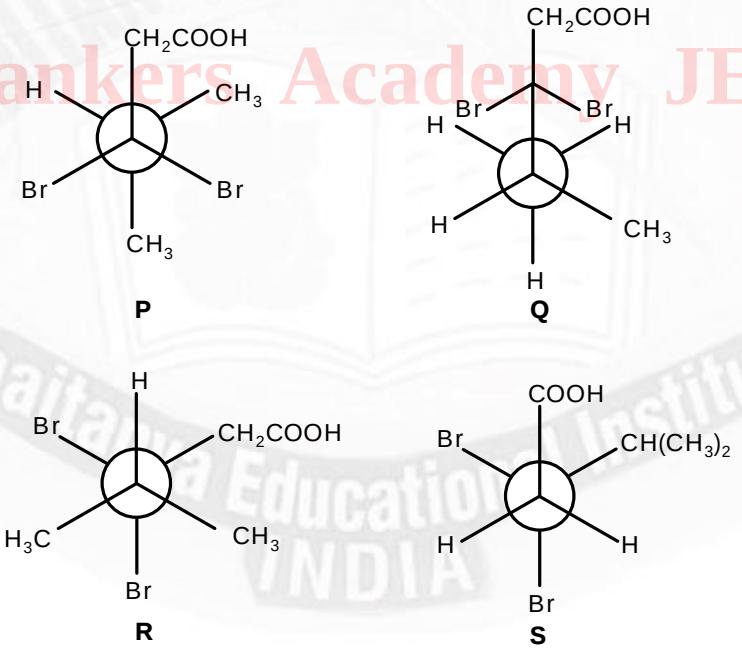


Column I has the structures of isomers of D-Threose and Column II has the relation of structure given in Column I with D-Threose. Choose the option(s) with correct relation.

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	Column I	Column II (Relation with D-Threose)
A)		Identical
B)		Enantiomer
C)		Diastereomer
D)		Enantiomer

28. Newman projections of **P, Q, R** and **S** are shown below:



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Which of the following pairs represent(s) identical molecules?

A) **P** and **Q** B) **Q** and **R** C) **R** and **S** D) **P** and **S**

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29. The correct statement(s) is(are):

A) For two gases at the same temperature and pressure, the gas of smaller molecular dimensions travels larger distance between successive collisions.

B) For a container of fixed volume, the distance between collisions is determined by the number of molecules present in the given volume, not by the speed at which they travel.

C) Real gases tend to behave ideally when average distance travelled by molecules between successive collisions is much larger than molecular diameter.

D) At constant volume the mean free path varies inversely with absolute temperature.

30. Consider a hydrogen-like atom with nuclear charge Z. The electron is in a stationary state.Which of the following statement(s) is(are)**correct**?

A) For the ground state, the probability density of finding the electron at the nucleus is maximum.

B) For the ground state, the radial probability distribution has exactly one maximum.

C)For a given principal quantum number n, the number of radial nodes is independent of the nuclear charge Z.

D) If the nuclear charge is increased from $Z$to $2Z$, the distance at which the radial probability distribution is maximum becomes half of its original value, for the ground state.

SECTION – III
(NUMERICAL VALUE TYPE)

• This section contains **SIX** (06) questions. The answer to each question is a **NUMERICAL VALUE.**
• For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer. **If** the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.
• Answer to each question will be evaluated **according to the following marking scheme**:
Full Marks: +4 **If** ONLY the correct numerical value is entered;
Zero Marks: 0 In all other cases

31. How many of the following are paramagnetic in nature

$VO^{+2}, Cu^{+}, I_2, I_2^{-}, Cu^{+2}, O_3^{-}, O_2, S_2$

32. How many of the following complexes are involving sp^3d^2 hybridization.

$\left[Co(NO_2)_6\right]^{-3}, \left[Fe(CN)_6\right]^{-3}, \left[Ni(NH_3)_6\right]^{+2}, \left[Co(H_2O)_6\right]^{+2}, \left[Cu(H_2O)_6\right]^{+2}$

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33. Consider the following reaction sequence:

$$\text{PhCOOEt} \xrightarrow[2)\,NH_4Cl(aq)]{1)\,MeMgBr(excess)} P \xrightarrow[\Delta]{Conc.H_2SO_4} Q \xrightarrow[2)\,Zn/H_2O]{1)\,O_3} R + S$$

(1 mol)

$$R + 4S \xrightarrow{NaOH} T$$

$$T + 2R \xrightarrow{\text{dry HCl}} U$$

Considering only the amounts of R and S formed in the above sequence, the maximum mass of **U** that can be produced is_____.

Given H = 1, C = 12, O = 16.

34. A certain aliphatic carboxylic acid $\mathbf{A}(C_4H_8O_2)$ is treated with $NaOH$ to form salt **B**. An aqueous solution containing 176 grams of **B** is subjected to electrolysis using platinum electrodes. At the anode, a hydrocarbon **C** is the primary organic product. The electrolysis process operates at a current efficiency of 80%. Compound **C** is passed over a $Cr_2O_3 / A\ell_2O_3$ catalyst at $500^\circ C$ to yield compound **D** with a yield of 75%. Compound **D** is then treated with a mixture of concentrated HNO_3 / H_2SO_4, and the resulting product is reacted with Sn/HCl followed by aqueous $NaOH$ to give compound **E**. The overall yield from **D** to **E** is 50%. Compound **E** is subsequently divided into two equal portions. The first portion is treated with concentrated H_2SO_4 and heated at $180^\circ C$ to form **F** with a yield of 90%. The second portion is reacted with excess CH_3I in the presence of Na_2CO_3 to form **G** with a yield of 80%. Finally, **F** is reacted with $NaNO_2$ and $HC\ell$ at 273–278 K, and the resulting mixture is added to **G** in a buffered medium to produce product **P**. Sodium salt of this final product is a common laboratory reagent that exhibits a distinct color change in the pH range of 3.1–4.4.

Data Provided:

Atomic masses: H = 1, C = 12, N = 14, O = 16, S = 32, Na = 23.

The final mass in grams of the sodium salt of product **P** obtained is_____.

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35. $NaC\ell$ crystallizes in the rock-salt structure in which $C\ell^-$ ions form an FCC lattice and Na^+ ions occupy all octahedral voids. A sample of NaCl contains Schottky defects such that 10 out of every 100 $NaC\ell$ formula units are missing. It is further doped with 2 mol% $MgC\ell_2$. The density of the crystal is $1.63\,g\,cm^{-3}$.

Assume that there is no distortion in the crystal structure due to defects.

Find the shortest distance (in pm) between Na^+ and $C\ell^-$ ions in the crystal.

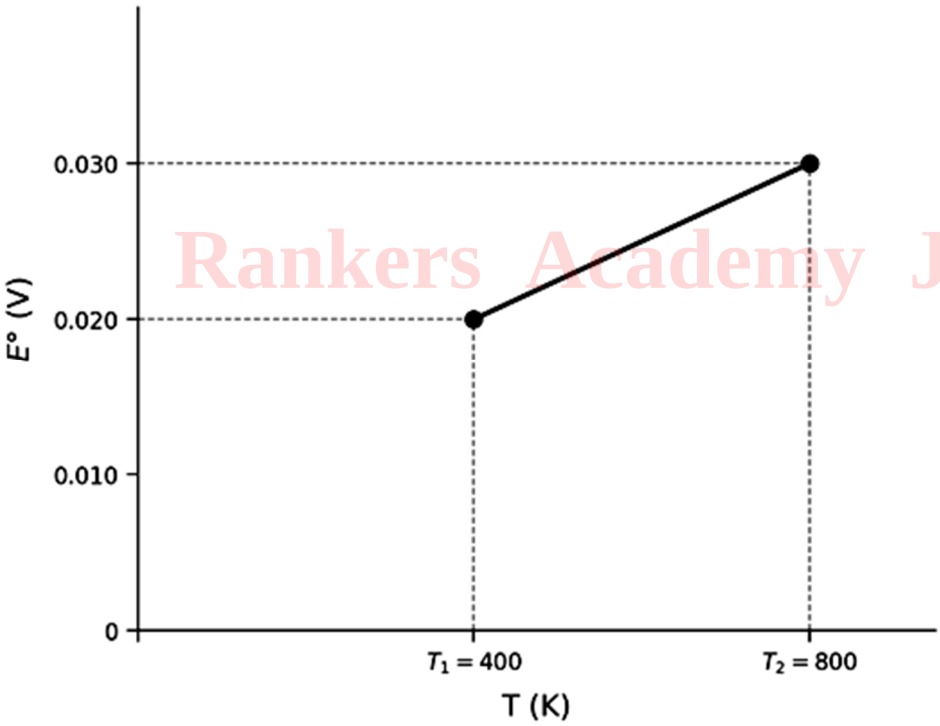
Given: $N_A = 6\times10^{23} mol^{-1}$

Atomic masses: $Na = 23$, $Mg = 24$, $C\ell = 35$

36. A galvanic cell operates using the reaction:

$A^{2+}(aq) + B(s) \longrightarrow A(s) + B^{3+}(aq)$

where all species at their standard states. The standard EMF of the cell (E°) is measured at two temperatures T_1 and T_2, and the variation is shown in the figure below:



Given:

F = 96500 C/mol

R = 2cal/mol·K

Find the magnitude of $\Delta H°$ in kJ/mol.

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MATHEMATICS **Max Marks: 66**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains **SIX** (06) questions.
• Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
• For each question, choose the option corresponding to the correct answer.
• Answer to each question will be evaluated **according to the following marking scheme:**
Full Marks: +3 If ONLY the correct option is chosen;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: -1 In all other cases

37. The distance covered by a particle moving in a straight line from a fixed point is 's'. Where $s^2 = at^2 + 2bt + c$, if t is time and a, b and c are constants then the acceleration is $\left(b^2 \neq ac\right)$

A) Proportional to square of distance
B) Inversely proportional to cube of speed
C) Inversely proportional to cube of distance
D) Proportional to square of speed

38. A circle is described upon AA' the major axis of ellipse as diameter. Let P be any point on circle. Let AP and $A'P$ are joined cutting the ellipse in Q and Q' respectively. If $\frac{AP}{AQ} + \frac{A'P}{A'Q'} = 3$, then the value of $\frac{1}{e^2}$ is ('e' is eccentricity of ellipse)

A) 2 B) 3 C) 4 D) 5

39. If z is any complex number lying on the curve $C: \left|z^2 - 6\right| = 6$, and $a = \min\left(\sum_{r=0}^{4}\sum_{k=0}^{2021}\left|ze^{\frac{ik\pi}{3}} - \frac{e^{ir\pi}}{1011}\right|\right)$, (where i (iota) = $\sqrt{-1}$) then number of positive integral divisors of 'a' are equal to

A) 3 B) 4 C) 5 D) 6

40. n biased coins, with m^{th} coin having probability of throwing tail equal to $\frac{1}{2m+1}\left(m = 1, 2, ..., n\right)$ are tossed once. The probability an odd number of tails, if results for each coin are independent, is $\frac{p}{q}$, when $n = 13$, find $p + q$. (p, q are coprime natural numbers)

A) 40 B) 37 C) 33 D) 29

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41. Consider a tetrahedron dice that has four integers 1, 2, 3, 4 written on its faces. Roll the dice 2000 times and for each i = 1, 2, 3, 4. Let f(i) represents the number of times i written on the bottom face. Let A denote total sum of numbers on the bottom face for these 2000 rolls. If $A^4 = 6144\ f(1)\ f(2)\ f(3)\ f(4)$ and $g(x)$ is a polynomial such that $g(x)=0$ has all real roots. If $a,b,c,d \in R$ and let,

$g(x)=x^4+(f(1)-d)x^3+(f(2)-c)x^2+(f(3)-b)x+(f(4)-a)$. Given $|g(i)|=1$, (where i (iota) = $\sqrt{-1}$). Then, which of the following is **INCORRECT**?

A) $\left|\frac{f(1)-f(2)}{f(4)}\right|=2$ B) $\frac{d}{240}=6$ C) $g(-2)=16$ D) $\frac{a}{20}=12$

42. Identify the number of **CORRECT** statement(s)

1) Let $f:[a,b]\to[a,b]$ be a differentiable function with $f(a)=b$ and $f(b)=a$ then there exist $c,d\in(a,b)$ such that $f'(c)\cdot f'(d)=1(c\neq d)$

2) Let $f(x)$ is a differentiable function in $[a,b]$ then there exist $c_1,c_2\in(a,b)$ such that $\frac{f(b)-f(a)}{b-a}=\frac{f'(c_1)+f'(c_2)}{2}$

3) If $f''(x)<0$ and $a<b<c$ then $(c-a)f(b)+(a-b)f(c)+(b-c)f(a)<0$

4) $\lim\limits_{x\to0^+}\left(\frac{x^{x^{x^x}}-x^{x^x}}{x^x-x}\right)=1$

A) 1 B) 2 C) 3 D) 4

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

- This section contains **SIX** (06) questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated **according to the following marking scheme**:

Full Marks: +4 If only (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but ONLY three options are chosen;
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Partial Marks: +1 If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: -2 In all other cases

43. ΔABC is an equilateral triangle with circumcentre (1, 2) and vertex A lying on the line $5x+2y=4$. A circle with centre I(2, 0) passes through the vertices B and C and intersects the sides AC and AB and D and E respectively. If area of quadrilateral BCDE is Δ, then

A) $AB=5\sqrt{15}$ B) $AB=6\sqrt{15}$ C) $\Delta=90\sqrt{3}$ D) $\Delta=60\sqrt{3}$

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44. Let $f:[0,1]\to R$ be defined as $f(x)=h(x)+9x-4$, where $h(x)$ is some integrable function. If $\int_0^1 f(x)dx=\int_0^1 x\,f(x)dx=\frac{1}{2}$, then the possible value(s) of $\int_0^1\left(f^2(x)+\left(x^2-x\right)h'(x)\right)dx$ can be

A) $\frac{5}{2}$ B) $\frac{3}{2}$ C) 1 D) 2

45. A football is played according to following rules:

(i) Each team plays exactly one game against each of the other teams.

(ii) Winning team get and 1 and losing team gets no point.

(iii) For a draw both teams get 0.5 points.

After the league was over, the teams were ranked according to their number of points. Following observations were noted.

I) Exactly half of the points earned by each team were against the 10 teams which finished at the bottom of the table.

II) Each of the bottom 10 team earned half of their total points against the other nine teams in bottom ten.

Which of the following statement(s) is/are **CORRECT**?

A) Total number of teams playing in tournament is 25.

B) Total number of teams playing in tournament is 30.

C) Number of matches between bottom 10 and remaining is 150.

D) Number of matches between bottom 10 and remaining is 200.

46. Let the conics $x^2=xy+1$ and $xy=x^2+1$ have eccentricities e_1 and e_2 respectively then:

A) $e_1>e_2$ B) $e_1<e_2$ C) $e_1^2+e_2^2=8$ D) $e_1^{-2}+e_2^{-2}=1$

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47. Consider, $f(x)=x-\sin x$, $g(x)=\cos^{-1}\left(e^{-\frac{x^4}{2}}\right)$ and $h(x)=\tan^{-1}\left(\frac{2x}{1+x^2}\right)$ then which of the following is/are **CORRECT**?

A) $\lim\limits_{x\to 0}\frac{f(x)-g(x)}{x^2}$ is equal to $\frac{1}{2}$ B) $\lim\limits_{x\to 0}\frac{g(x)-xh(x)}{x^2}$ is equal to –1

C) $\lim\limits_{x\to 0}\frac{f(2x)-h\left(x^3\right)}{x^3}$ is equal to $-\frac{2}{3}$ D) $\lim\limits_{x\to\infty}\frac{\cot^{-1}\left(h(x)\right)-2g(x)}{\cos^{-1}\left(x\left(1-\cos\frac{1}{x}\right)\right)}$ is equal to –1

48. Which of the following is/are **CORRECT**?

A) The value of $\sum\limits_{n=1}^{10}\frac{15^n}{\left(5^n-3^n\right)\left(5^{n+1}-3^{n+1}\right)}$ is $\frac{1}{2}\left(\frac{3}{2}-\frac{3^{11}}{5^{11}-3^{11}}\right)$

B) If $x,y>0$ such that $\frac{1}{x}+\frac{1}{y}=1$, then minimum value of $\frac{31^x}{x}+\frac{65^y}{y}$ is 2015

C) If $S_n=\frac{n\left(2n^2+9n+13\right)}{6}$ and $t_1+t_2+t_3+.....+t_n=S_n$ then the value of $\sum\limits_{r=1}^{\infty}\frac{1}{r\sqrt{t_r}}$ is 1

D) If $(b-c)x^2+(c-a)xy+(a-b)y^2$ is a perfect square then a,b,c are in A.P.

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SECTION – III
(NUMERICAL VALUE TYPE)

• This section contains **SIX** (06) questions. The answer to each question is a **NUMERICAL VALUE.**
• For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer. **If** the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.
• Answer to each question will be evaluated **according to the following marking scheme**:
Full Marks: +4 **If** ONLY the correct numerical value is entered;
Zero Marks: 0 In all other cases

49. Let $y=f(x)$ defined in [0, 2] satisfies the differential equation $y^3y''+1=0$ where $f(x)\geq 0\forall x\in D_f$ and $f'(1)=0, f(1)=1$ then find the maximum value of $f(x)$.

[Note: D_f denotes the domain of the function and y'' denotes the 2nd derivative of y w.r.t. x].

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50. If a function $f(x)$, such that $f:[0,1]\to R$ satisfy relation

$$3\left(\int_0^1 f(x)\cdot dx-\int_0^1 f^2\left(x^2\right)\cdot dx\right)=1$$ then find value of $2\cdot f^2\left(\frac{1}{5}\right)+3\cdot f^2\left(\frac{1}{10}\right)$.

51. If G is geometric mean of binomial coefficients in the expansion of $(1+x)^n, n\in N$, then find $\left[\lim_{n\to\infty} G^{1/n}\right]$ (where [.] is greatest integer function)

52. The value of $1+\sum_{k=1}^{8}\frac{\cos k\theta}{\cos^k\theta}$ for $\theta=30^\circ$ is p, then $\frac{\sin 9\theta}{p\sin\frac{\pi}{6}\left(\cos\frac{\pi}{6}\right)^8}$ is equal to

53. Number of all positive real numbers $\alpha\le 64$, such that $[\alpha]+[2\alpha]+[3\alpha]+....+[64\alpha]$ is divisible by 64 is λ. Also number of pairs $(a,b);a,b\in N$, which satisfy equation $\left(a^2+2b^2\right)^2+8b^2=65$ is μ. Find $\lambda+\mu$.

(Where [.] represents greatest integer function)

54. A person has 6 cards A, K, 2, Q, 10, 9. The person randomly draws the cards one by one with replacement till he gets 3 consecutive A. If P_n represents the probability that atleast n cards are drawn, then find value of $216\cdot P_{11}-5\cdot P_8-30\cdot P_9-180\cdot P_{10}$.

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